

Design, Implementation of an NDPA-Compliant IoT Edge Vulnerability Assessment System

Abstract

With the increasing use of Internet of Things (IoT) devices in industries across Nigeria, the threat to cybersecurity and data privacy compliance is becoming a challenge for the industries. The Nigeria Data Protection Act (NDPA) 2023 mandates organisations to put in place mitigations to secure personal data processed via connected systems, but there is no vulnerability assessment tool which directly maps IoT security vulnerabilities with the requirements of NDPA. The design, implementation, and empirical testing of an edge vulnerability assessment system for IoT devices, compliant with NDPA, have been presented in this paper, using Python, Flask, Nmap, SQLite, and ReportLab. Combined in one single platform, the system comes with four features: multi-method device discovery, rule-based vulnerability detection across 18 IoT-relevant ports, privacy risk assessment reference to 6 NDPA data protection principles, and automatic compliance report generation. The four devices used for empirical evaluation on a live MiFi network included a MiFi router and an Android-based car head-unit, and resulted in one vulnerability finding (medium severity) correctly mapped to NDPA Section 24(1)(b), six privacy risk findings mapped to Sections 10, 24(1)(b), 24(2), and 25, and a network privacy score of 3.1/10. All 5 test cases have passed. The system is designed in a stand-alone application of the Windows executable that does not

need any particular expertise to configure, thus enabling Nigerian organisations of all sizes to have their IoT compliance assessed by non-specialist compliance officers.

Keywords: *Internet of Things, IoT security, vulnerability assessment, Nigeria Data Protection Act, NDPA compliance, edge computing privacy risk assessment, network scanning.*

1. Introduction

The Internet of Things has evolved beyond the speed at which it can be secured. From hospital equipment to vehicle infotainment, billions of smart devices gather and transmit information through interwoven networks (Obaidat et al., 2020). The digital economy is fostering the proliferation of connected devices and edge computing systems in Nigeria, which is driving widespread adoption of IoT in banking, healthcare, logistics, and smart infrastructure (NDPC, 2024). Edge computing, which involves processing data near the source, can reduce latency and improve real-time responses, albeit creating a larger attack surface, and increasing the number of security vulnerabilities from centralised systems (Alwarafy et al., 2020; Roman et al., 2021).

The production of most IoT devices lacks security mechanisms, making them susceptible to attacks of unauthorised access, weak authentication, insecure communication protocols, data interception and privacy breach (Anand et al., 2020; Tawalbeh et al., 2020). Organisations now have concrete legal obligations to put in place technical and organisational measures to protect personal data that is processed by these systems under the Nigeria Data Protection Act (NDPA) 2023. The broader societal impact is a loss of public confidence in the digital

infrastructure and the financial, legal and reputational implications for non-compliance (Iyamu & Mhlongo, 2022).

Nevertheless, there is one significant disadvantage: There is currently no vulnerability assessment tool that directly maps the vulnerabilities identified in IoT to the NDPA requirements. Technical scanning tools such as EXPLIoT, IoT Inspector (NYU, 2021), OWASP IoT Top 10 (2023), and NIST NISTIR 8259 (2020) are focused on international frameworks such as GDPR, ISO/IEC 27001 and NIST standards, which are not localised to Nigeria's regulatory requirements. There is consequently no practical tool available to Nigerian organisations to systematically evaluate their IoT infrastructure in terms of whether they comply with the NDPA (Greenleaf, 2021; Voigt & Von dem Bussche, 2021).

This paper fills that gap by proposing, designing, implementing and evaluating an empirical system for evaluating vulnerabilities at the edge of the IoT. The system is built in an iterative progression over five prototype phases, with the use of Python, Flask, Nmap, SQLite, and ReportLab and has been tested using a live scan of a real-world MiFi network. The main contributions of this work are: (1) a structured three-stage compliance mapping framework for the compliance of the IoT from the NDPA 2023, (2) a two-layer IoT device identification system, (3) an integrated platform for vulnerability and privacy assessment packaged into a standalone executable for non-specialist deployment, and (4) empirical evidence showing connected automotive devices as an overlooked vulnerability and privacy assessment risk category for NDPA compliance.

The rest of this paper is organized as follows. A review of related work is provided in Section 2. The methodology is detailed in Section 3. The system design and architecture are presented in Section 4. The empirical results and evaluation are presented in Section 5. The final section, Section 6, presents suggestions for future research.

2. Literature Review

2.1 IoT Security Vulnerability and Edge Computing Risks

The main features of IoT devices are limited resources, different architectures, and an older focus on functionality than on security (Anand et al., 2020). The OWASP IoT Top 10 (2023) lists the most common group of vulnerability classes: weak passwords, insecure network services, outdated firmware, lack of secure update mechanisms, and use of insecure or obsolete protocols. NIST NISTIR 8259 (2020) offers general information for device manufacturers to consider when it comes to securing devices from the outset. However, edge computing exacerbates these threats as it expands the attack surface with multiple nodes having different security postures compared to the centralized cloud architectures (Shi et al., 2020; Roman et al., 2021). Radanliev research shows that edge environments are among the most vulnerable environments for cascading compromise, with a single vulnerable gateway potentially compromising all devices on the network.

2.2 Privacy Compliance and the NDPA 2023

The Nigeria Data Protection Act, 2023, is the biggest change in data governance in Nigeria since the repealed NDPR. It sets out seven data protection principles – lawful processing (Section 10), confidentiality and access control (Section 24(1)(a)), security of processing (Section 24(1)(b)), data minimisation (Section 24(1)(c)), storage limitation (Section 24(1)(d)), data protection by design (Section 24(2)), and breach notification (Section 25), and requires organisations to put in place technical measures proportionate to the data privacy risks of the processing activities. The NDPA's applicability to IoT deployments is clear as it is explicitly written so as to include automated processing systems such as connected devices.

Iyamu and Mhlongo (2022) highlight structural challenges to cybersecurity compliance in developing economies and point out that tools need to be locally contextualised, which is a major hurdle in the practical implementation of cybersecurity.

2.3 Existing Assessment Tools and Their Limitations

A survey of current IoT assessment tools shows that the existing ones are technically capable but not localised with respect to regulatory requirements. At the enterprise level EXPLIoT offers firmware analysis, network scanning and vulnerability identification, but it is a big investment in infrastructure and doesn't provide any NDPA mapping. The most methodologically comparable tool to the present work is IoT Inspector (NYU, 2021), which tracks device traffic flows to identify privacy-threatening behaviors, but does not actively scan for vulnerabilities or produce any compliance results that reference any data protection framework. OWASP IoT Top 10 and NIST NISTIR 8259 are not tools to run, but rather frameworks that offer guidance but not the tools for assessment. However, Zhang et al. (2020) and Lu and Xu (2020) have proposed privacy-preserving architectures for the edge environment without considering the compliance reporting obligations of national data protection laws.

The three critical gaps found in all the tools reviewed are: the absence of a tool that correlates IoT security findings with the NDPA 2023; the fact that there is no lightweight tool that is accessible to non-specialist organisations within a developing economy environment; and the absence of a tool that combines vulnerability assessment and privacy compliance assessment within a single platform. All three gaps are covered in the present work.

3. Methodology

3.1 Research Design

This study is a pragmatist type of research mixed method. The qualitative analysis of NDPA 2023 was conducted systematically to determine the compliance requirements to be considered by an IoT assessment system, and literature reviewing was conducted to find out the gap which the IoT assessment system would fill. From a quantitative point of view, the system implemented was tested in a real (empirical) manner by performing live network scans and comparing their results with the knowledge database of the test environment. The study is located in the Design Science Research paradigm (Hevner et al., 2004), in which the designed and evaluated artefact, the NDPA-compliant scanning system, will be the main research contribution not theoretical knowledge only.

3.2 System Development Methodology

The system development methodology was chosen to be the Prototyping Model. This is because three limitations exist: compliance mapping could not be fully detailed prior to project development, because important decisions regarding NDPA provision applicability could be determined only through a process of iterative testing; the project comprised of a single academic research endeavour, and did not have the team structure typical of Agile methodologies; and it was an explicitly prototype project rather than a production system itself (Pressman & Maxim, 2020). The development was conducted in five steps: (P1) system architecture and database foundation; (P2) device discovery and port scanning engine; (P3) vulnerability analysis engine, and NDPA mapping; (P4) privacy risk assessment module; and (P5) reporting dashboard and executable packaging.

3.3 NDPA Compliance Mapping Framework

The NDPA compliance mapping methodology, the main theoretical contribution of this research, was developed in a systematic three step process. First, each vulnerability or privacy risk detected was categorized by a name (Unencrypted Communication, Unknown Device). Secondly, the category was linked to the specific part of NDPA, which is implied by its technical features and its obligations are defined in the text of the Act. Third, an explanation in plain language was provided in a statement to outline the legal requirement in a way that a non-specialist compliance officer could understand. This framework results in 7 vulnerability-to-NDPA mapping and 6 principle-to-NDPA mapping, which cover Sections 10 and 24(1)(a) – 24(1)(d) and 24(2) and 25 of the Act.

3.4 Evaluation Approach

The system was tested using a black-box functional testing approach using live network validation with ground truth on a 192.168.0.0/24 MiFi subnet on 29 May 2026 (Session 15). A set of five test cases were identified, including device discovery, port scanning, vulnerability analysis, privacy assessment and report generation. The researcher personally knew about the four devices which were linked through the network and hence was able to provide the ground truth. Criteria were technical accuracy, correctness of the mapping in the NDPA, and objective attainment.

4. System Design

4.1 Architecture

The system consists of a three-tiered client-server system. It uses Chart.js data visualisations and a browser-based interface via a Flask template engine called Jinja2. This consists of Flask route blueprints and four separate modules: network scanning engine, vulnerability analysis engine, privacy risk assessment engine, and PDF report generator. The data layer contains five entity types: ScanSession, Device, OpenPort, Vulnerability and PrivacyRisk, with one-to-many relationships which allow complete tracing from finding to session. It is installed as a standalone Windows executable file using PyInstaller, and only Nmap is an external system-level dependency.

4.2 Device Discovery and Two-Layer Identification

This method of host discovery involves sending ARP ping scans (-PR) and TCP SYN ping on the ports 22, 80 and 443 (-PS22,80,443), all with a timeout of 10 seconds per host. A combination which reliably finds devices on MiFi and firewalled networks on which ICMP ping is denied. Port scanning, using Nmap's service version detection (-sV) to perform against 18 ports per host found, with a 60-second per host time limit to prevent worst case scan time.

There is a two-layer system of device type identification. Layer 1: Fingerprints port-based against eight IoT device profiles, each with a minimum match threshold to avoid false positives from shared ports. On Layer 2a, the hostname pattern matching function can be used to match 22 standard patterns supported by manufacturers, such as Android hostname pattern (android-[0-9a-f]+\$), Apple device patterns and Windows machine name patterns,

allowing the support of consumer and embedded devices that don't offer any IoT-specific ports. Layer 2b – MAC address OUI lookup is a last resort for 10 OUI groups.

4.3 Vulnerability Analysis Engine

The vulnerability analysis engine uses a rule set that is based on a dictionary of seven detection rules for the most common vulnerabilities exposed in open port enumeration, including Telnet (port 23), FTP (port 21), unencrypted MQTT (port 1883), HTTP administrative interfaces (ports 80/8080), VNC remote desktop (port 5900), RTSP streaming (port 554) and TR-069 router management (port 7547). The trigger ports, severity (critical/high/medium/low), and the `ndpa_category` key that connects each rule to the `NDPA_MAPPINGS` dictionary are also specified. This separation allows for the evolution of NDPA provision references without changing detection logic. The device security scores are based on a weighted algorithm: critical = 4 points, high = 2.5 points, medium = 1 point, and normalised to 10 points.

4.4 Privacy Risk Assessment Engine

The privacy risk assessment engine contains seven rules which are mapped to six NDPA data protection principles. Privacy rules are not vulnerability related, and they can be based on device type or port profile—or both—and make findings that fire on devices that have no open ports, if the device type only indicates a compliance issue. Table 1 reflects the NDPA compliance mapping framework by both engines.

Table 1: NDPA Compliance Mapping Framework

NDPA Provision	Principle	System Compliance Requirement
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Section 10	Lawful Processing	Flag devices enabling unlawful data collection or processing
Section 24(1)(a)	Confidentiality & Access	Identify absent or weak authentication mechanisms on IoT devices
Section 24(1)(b)	Security of Processing	Detect unencrypted protocols and insecure data transmission practices
Section 24(1)(c)	Data Minimisation	Flag devices collecting data beyond what is necessary for stated purpose
Section 24(1)(d)	Storage Limitation	Identify NAS/storage devices; flag need for data retention policy review
Section 24(2)	Data Protection by Design	Flag unnecessarily exposed services and unidentified devices (accountability)
Section 25	Breach Notification	Flag vulnerabilities that, if exploited, would trigger breach reporting obligations

4.5 Reporting Layer

The reporting layer offers 3 output formats. The aggregate dashboard displays vulnerability severity distributions and vulnerability per-session scan history using Chart.js doughnut and bar charts. Web reports, produced per session, display all the device, port, vulnerability, and privacy risk findings in a structured browser interface, including references to the NDPA provisions. On demand: PDF reports are generated with ReportLab's Platypus document engine, streaming to the browser, as a file download, without the need to write to a disk.

path-handling issues are fixed in the standalone .exe entry point (launcher.py); the working directory is set to make sure that the SQLite database exists with the executable, and the browser interface is opened automatically when the launcher.py is executed.

5. Results and Evaluation

The Test Environment and Session 15 overview will be given. The Test Environment and Session 15 overview will be provided.

The evaluation was conducted on a live MiFi mobile hotspot network (192.168.0.0/24) on 29 May 2026, comprising four devices: a MiFi router/gateway (192.168.0.1), an Android-based car head-unit (192.168.0.133, hostname: android-bcbd3015cd2b640d), and two other connected devices at 192.168.0.134 and 192.168.0.148 (hostname: ORION). An Android car head-unit (in-vehicle infotainment system running Android) is one form of IoT edge device that is becoming a common feature in Nigeria and is not normally considered in traditional IT risk assessments but can capture the history of your location, driving habits and personal details from synched mobile devices through Bluetooth connectivity. The Session 15 executive summary is included in Table 2.

Table 2: Session 15 Executive Summary

Metric	Value
Scan Status	Completed
Scan Duration	~7 minutes
Devices Discovered	4

Open Ports Found	3
Total Vulnerabilities	1 (Medium)
Privacy Risks Found	6 (1 Critical, 1 High, 4 Medium)
Network Privacy Score	3.1 / 10
Test Cases Passed	5 / 5

5.2 Device Discovery and Identification

All four devices were discovered successfully within the 7 minute scan window, which proves that the multi-method ARP and TCP SYN device discovery approach works well for enumerating devices on firewalled MiFi networks. Both the MiFi router and the car head-unit were correctly identified by the two-layer identification system: Router/Gateway (Layer 1, port fingerprinting) and Android Device (Layer 2a, hostname pattern matching against android-[hexstring]). The devices at 192.168.0.134 and 192.168.0.148 (orion) were both returned correctly as Unknown Device — orion is a user-assigned machine name in an unformatted application layer MAC address structure and neither device returned an accessible prefix in the application layer. This is the right answer: If there is not enough evidence, then the system does not make an inference about the type of the device.

5.3 Vulnerability Findings

The vulnerability engine produced one medium vulnerability: Unencrypted HTTP administrative interface on router (port 80, MiWiFi HTTP Service) (NDPA Section 24(1)(b))

and Section 25). The security risk score of the router was 1.0/10. None of the three other devices exposed vulnerability-triggering ports, which gave them a score of 0.0/10. Of course, the vulnerability results for consumer and embedded devices were not IoT specific, so they don't reveal anything about the devices, but that's why the complementary privacy risk assessment layer is necessary.

5.4 Privacy Risk Findings

The privacy engine identified six privacy issues on four devices. The router was found for three findings: a critical finding for Unencrypted Data Transmission (Section 24(1)(b)) and a high finding for Video Surveillance Device Detected (Section 10), which is a known false positive that is explained below, and a medium finding for Router/Gateway accountability (Section 24(2)). The remaining three devices were each issued a medium (24(2)) Unknown Device accountability finding. The car head-unit's accountability finding is especially noteworthy, since the device isn't vulnerable to a port-based attack, but the NDPA's accountability obligation under Section 24(2) still extends to it: The organisation that is using a car head-unit that collects location data, driving behaviour and contact details via Bluetooth still has to document and explain the processing activity, even if the car head-unit is not vulnerable to a port-based attack.

The finding from the video surveillance in the router is a true positive, but it is false positive because it will hit on port 80, which is common to both routers and IP cameras. As mentioned in Section 6, this design restriction necessitates the need to qualify the device type in the rule allowing it to trigger only when the device is also qualified as IP Camera.

5.5 Objectives Attainment

Table 3: Research Objectives Attainment Summary

Obj.	Objective	Assessment
O1	Create a model to evaluate vulnerabilities in IoT edge infrastructure	Fully Met — 7 rules, correct NDPA mapping, proportionate scoring
O2	Create a system to identify IoT devices and common security weaknesses	Substantially Met — router and car head-unit identified; 2 devices Unknown
O3	Implement a privacy risk assessment module	Fully Met — 6 findings across 4 devices, all with NDPA principle mapping
O4	Integrate NDPA compliance mapping mechanisms	Fully Met — all 7 findings mapped to named NDPA provisions (verified)
O5	Create a reporting dashboard producing compliance reports	Fully Met — web report, PDF export, Chart.js dashboard all operational

5.6 Comparison with IoT Inspector (NYU, 2021)

The closest, preceding work is IoT Inspector, an empirically tested, network-based privacy evaluation tool, which is the most methodologically similar research. Both systems use MAC OUI and hostname analysis to identify devices. But the systems are placed in different places in the assessment arena. Continuous passive traffic monitoring is the strength of IoT Inspector, as it is great to find data leakage to third party endpoints or unexpected traffic to/from devices that cannot be found by port scanning that reveals privacy-concerning behaviours. The present system would be effective in detecting vulnerabilities (active) and, importantly, mapping compliance with the NDPA (NDPA-specific): IoT Inspector would see the unencrypted HTTP traffic on the router but would be unable to detect the VULN-004 vulnerability, assign a medium rating, reference Section 24(1)(b) of the NDPA, or make a recommendation for remediation. IoT Inspector does not report any findings to a regulatory framework. Each of the seven findings in Session 15 of the present system is connected to a named NDPA provision, the ability that has been missing from any other tool for Nigerian organisations. The active vulnerability scanning and passive traffic monitoring are complementary; the combination of the two would yield the most complete NDPA compliance posture assessment.

6. Conclusion

In this paper, the design, implementation and empirical testing of an NDPA-compliant IoT edge vulnerability assessment system has been presented and its feasibility to develop a lightweight, locally contextualised IoT security and compliance assessment system has been demonstrated within the scope of a single academic prototype. The system fills the major gap in the literature, which lacks tools that map IoT security findings to the obligations in the

NDPA 2023, by providing a structured three-stage compliance mapping framework based on seven categories of vulnerability and six categories of data protection in Section 10, 24, and 25 of the NDPA 2023. Four out of five research objectives were completely successful and one was substantially successful, with the system correctly identifying an Android car head-unit as an IoT edge device with NDPA accountability obligations, which many of the existing tools that were reviewed were missing of this device category.

Four constraints are admitted. First, the prototype was tested on four devices, but performance at enterprise scale has yet to be validated. Second, the rule-based passive scanning method is unable to detect vulnerabilities in the firmware, zero-day threats or vulnerabilities in the application layer. Third, privacy findings are not verified by actively inspecting the data flow, but rather by inferring it from the device type and port profile. Fourth, the NDPA compliance mappings have not been endorsed by a qualified Nigerian data protection practitioner.

The following seven recommendations are put forward for future development (in order of priority): (R1) Further work on the video surveillance privacy rule to qualify devices, removing the confirmed false positive; (R2) Development of dedicated privacy rules for Android car head-units and mobile devices; (R3) Add device-type matching to existing privacy rules, to provide combined port and device-type matching; (R4) Require parallel scanning for improved performance on larger networks; (R5) Move to PostgreSQL for multi-user deployment and use of the rule; (R6) Extend NDPA coverage to include CBN, NCC, and National Health Act sector frameworks; and (R7) Add OS fingerprinting and mDNS service discovery to reduce Unknown Device classifications.

The overall impact of this research is the proposed method of localising tools for assessing the vulnerability of the Internet of Things at the national level, replicable in any country. The three-stage NDPA mapping approach is framework-agnostic, and could be applied to the Data

Protection Act for Kenya, the POPIA for South Africa, the Data Protection Act for Ghana or any other data protection instrument in Africa; it presents a model for the creation of compliance contextualised IoT security tooling across the continent.

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